Prince Agyei Tuffour

+1 6232733354 | prince.agyei.tuffour@gmail.com | GitHub

SUMMARY

Machine Learning Engineer with a Master's in Mathematics and proven experience in deploying ML models using TensorFlow and PyTorch. Successfully enhanced recommendation systems, boosting user engagement and improving predictive accuracy by up to 20%. Demonstrated strong cross-functional collaboration in research and development projects, optimizing model performance and accelerating feature rollouts.

SKILLS

- **Programming Languages**: Python, Javascript, C++, SQL, Java, Typescript
- **Tools & Technologies**: Git, Nextjs, Redis, AWS, Google Cloud, Playwright, Tailwind CSS, Keras, Pandas, ScikitLearn, Matplotlib, Deep Learning, Linux/Unix, PyTorch, Tensorflow, Selenium, Flask, Django, RESTful APIs, Express, PostgreSQL, MongoDB, React, SciPy, Streamlit, YOLO

WORK EXPERIENCE

dynaConnections Corporations **Mar 2025 - Present**

Software QA Engineer

- Designed and executed comprehensive test strategies—including manual and automated protocols, API testing, UI validation, and performance evaluations—to ensure compliance with industry standards.
- Achieved a 40% improvement in overall product reliability and a 60% reduction in post-deployment bugs by establishing robust testing frameworks and applying analytical problem solving.
- Developed detailed test cases and conducted regression testing, leveraging strong analytical skills to deliver deep insights into the software development lifecycle and enhance user experience.

Akomapa Health **Mar 2025 - Present**

Lead Software Developer/Executive Board Member

- Managed complete website workflow—including server maintenance, database optimization, and feature implementation—to support 500+ active student users monthly with 99.8% uptime, demonstrating proficiency in Python and C++.
- Engineered automated content delivery systems and implemented mobile-first design improvements that increased user engagement by 65% and boosted mobile traffic by 80%, while consistently writing clean, efficient, and sustainable code.
- Built an interactive community platform enabling students to access peer-reviewed health content, participate in wellness challenges, and connect with certified health professionals, showcasing cross-functional collaboration and strong communication skills.

Cita Marketplace.com **Jun 2022 - 2023**

Machine Learning Engineer - Intern

- Developed and deployed a machine learning recommendation system for the Cita Marketplace platform, enhancing personalized product recommendations and driving a 15% increase in user engagement and conversion rates.
- Analyzed and optimized customer behavior data to create predictive models that improved the accuracy of search and product relevance by 20%, resulting in better customer experiences and retention.
- Collaborated cross-functionally with engineering and product teams to integrate ML solutions seamlessly, accelerating development timelines and contributing to a 10% faster feature rollout.

Oregon State University **Sep 2021 - Dec 2023**

Graduate Research Assistant

- Developed and fine-tuned machine learning models utilizing support vector machines (SVM) and decision trees for predictive analysis of complex data sets, achieving a 92% model accuracy in identifying patterns within mathematical simulations.
- Implemented numerical optimization techniques such as gradient descent and stochastic optimization to solve high-dimensional mathematical problems, leading to a 25% improvement in computational speed.
- Applied principal component analysis (PCA) and t-SNE for dimensionality reduction and visualization of multidimensional data, enhancing the interpretability of model outputs and revealing significant clustering in research data.
- Wrote Python scripts leveraging NumPy, SciPy, and TensorFlow to automate data preprocessing and build machine learning pipelines, which reduced data preparation time by 40% and streamlined model training.

EDUCATION

Oregon State University, Corvallis OR

Master of Science, Mathematics

Kwame Nkrumah University of Science and Technology, Ghana

Bachelor of Science, Mathematics

PROJECTS

GradGPT.pro

- Designed and launched GradGPT.pro, an AI-powered graduate school assistant using GPT-4, streamlining school selection, application prep, and funding search for 1,500+ users.
- Integrated scraping tools, LLM-based Q&A, and secure chat saving to deliver tailored insights, boosting application efficiency and success for early-stage grad applicants.

Akomapa Health
- Engineered a Yale- and UCLA-backed health platform using React, Supabase, and TypeScript, reaching 25,000+ users and improving health literacy by 65% in rural Ghana.
- Achieved 80% rise in preventive care in high blood pressure and 300% outreach growth via offline features, volunteer medical training, and integrated emergency contact systems.

ByDay Jobs - Demo App
- Designed and built a location-based service platform (like Uber for professionals) using React, Next.js, Google Maps API, and Stripe to connect skilled workers with local demand in real-time.
- Currently in pilot testing with early users across 5 cities; features smart job matching, secure payments, and user reviews, laying the groundwork for scalable job creation and gig economy growth in underserved markets.

Stay Afloat - Customizable LMS for Ghanaian Institutions
- Developed a flexible SaaS learning platform tailored for Ghanaian institutions, enabling remote students to access quality online education with customizable branding and course structures.
- Empowered 12+ schools to expand online reach, achieving 60% higher course completion rates and 200% increase in student engagement through localized features, payment integration, and real-time progress tracking.

GPU Memory Profiler for TensorFlow & PyTorch - Open Source
- Contributed to GPU memory profiling tool optimizing deep learning workflows by visualizing model memory usage for TensorFlow and PyTorch.
- Enhanced debugging, training efficiency, and memory usage visualization by identifying bottlenecks, aiding various ML practitioners in improving model performance.